

Design Control-Traceability Matrix

Project: PDPROJ-2 (Cell-Guardian)

As Of: October 3, 2024 6:00 PM GMT

Matrix Density: Expanded

User Needs	Design Inputs	Design Outputs	Design Verifications	Design Validations
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Cell-Guardian

(UN-1) The user needs a device with a sensor that is able to continuously and accurately measure falls and the presence of a patient in their cell	(DI-1) The device shall incorporate a sensor board with an RF sensor (FMCW) for detection of inmate motion and falls	(DO-7) Assembly		
(UN-1) The user needs a device with a sensor that is able to continuously and accurately measure falls and the presence of a patient in their cell	(DI-2) The device shall incorporate an infrared camera to perform ambient temperature monitoring and possibly support movement detection	(DO-6) Thermal Camera		
(UN-1) The user needs a device with a sensor that is able to continuously and accurately measure falls and the presence of a patient in their cell	(DI-3) The device's sensor shall operate within a 5.8 GHz frequency range	(DO-7) Assembly		
(UN-1) The user needs a device with a sensor that is able to continuously and accurately measure falls and the presence of a patient in their cell	(DI-4) The device's sensors shall be functional within at least 13 feet to cover a 10x10x10 cell	(DO-7) Assembly		
(UN-1) The user needs a device with a sensor that is able to continuously and accurately measure falls and the presence of a patient in their cell	(DI-5) The devices sensors shall be powered by power over ethernet (POE) requiring no more than 10 Watts power consumption (currently 7.5W, POE =52W).	(DO-7) Assembly		
(UN-1) The user needs a device with a sensor that is able to continuously and accurately measure falls and the presence of a patient in their cell	(DI-7) The device shall incorporate a microprocessor\controller for the purpose of controlling the RF frontends and IR camera	(DO-5) RGB Camera		

(UN-1) The user needs a device with a sensor that is able to continuously and accurately measure falls and the presence of a patient in their cell _____	(DI-8) The microprocessor shall have the following minimum requirements: 1GB SDRAM, 8GB Flash Memory, 1.5GHz processor speed _____	(DO-7) Assembly _____		
(UN-3) The user needs to be able to view a visual image and perhaps a thermal image of the patient in addition to fall or alert notifications _____	(DI-9) The sensor board shall send data over an ethernet cable for display by a cloud front end _____	(DO-4) User Interface Wireframes _____ (DO-7) Assembly _____		
(UN-4) The user needs an unobtrusive and rugged enclosure that will protect all internal components from inmates and washing _____	(DI-10) The sensor shall be affixed within an enclosure no larger than the following dimensions: 10" x 10" x 3" _____	(DO-7) Assembly _____		
(UN-4) The user needs an unobtrusive and rugged enclosure that will protect all internal components from inmates and washing _____	(DI-11) The enclosure, with the affixed sensor board, shall not weigh more than 2 pounds. _____	(DO-7) Assembly _____		
(UN-4) The user needs an unobtrusive and rugged enclosure that will protect all internal components from inmates and washing _____	(DI-12) The enclosure shall be of the following material: polycarbonate _____	(DO-7) Assembly _____		
(UN-4) The user needs an unobtrusive and rugged enclosure that will protect all internal components from inmates and washing _____	(DI-13) The enclosure shall include ingress protection in accordance with IP66 requirements and fire rating of UL94 _____	(DO-7) Assembly _____		
(UN-4) The user needs an unobtrusive and rugged enclosure that will protect all internal components from inmates and washing _____	(DI-14) The enclosure shall connect to the POE on the top of the system near the ceiling mount to hide the connection _____	(DO-7) Assembly _____		
(UN-5) The user needs a device which can be mounted to a fixed surface _____	(DI-15) The mounting system shall be able to support the full weight of the device when mounted to either the wall or ceiling. _____	(DO-7) Assembly _____		

(UN-5) The user needs a device which can be mounted to a fixed surface	(DI-16) The device shall be able to withstand forces and impacts up to 5 Kg ball drop per IEC 60601-1.	(DO-7) Assembly		
(UN-6) The user needs a device that can withstand transportation, storage, and environmental conditions suitable for healthcare use.	(DI-17) Device will be operable under the environmental conditions below: Operating conditions: Temperature Range of +5 to +40 degrees C, relative humidity range of 15%-90%, and atmospheric pressure range of 700 hPa to 1060 hPa.	(DO-7) Assembly		
(UN-7) The user needs a device that provides alerts in response to falls or the absence of a prisoner	(DI-18) The device shall display an alert when it detects the absence of a prisoner in their cell	(DO-4) User Interface Wireframes		
(UN-7) The user needs a device that provides alerts in response to falls or the absence of a prisoner	(DI-19) The display shall display an alert when the device detects a fall	(DO-4) User Interface Wireframes		
(UN-7) The user needs a device that provides alerts in response to falls or the absence of a prisoner	(DI-20) The alerts shall be displayed on the Dashboard	(DO-4) User Interface Wireframes		
(UN-8) The user needs a device with appropriate labeling in compliance with applicable regulatory requirements.	(DI-21) Device labeling is compliant with FDA 21 CFR 801 and 21 CFR Part 1010 and includes the following information: Manufacturer Name and Address, Device Model, Description, Lot/Serial Number, Date of Manufacture, UDI, Rx Use Only Statement, and the electrical certification symbol.	(DO-2) Labelling		
(UN-8) The user needs a device with appropriate labeling in compliance with applicable regulatory requirements.	(DI-22) Device labeling is compliant with IEC 60601-1, IEC 60601-1-2, and IEC 60601-1-11 and includes the following information: IP66 rating symbol, IEC 60601-1 Table D.1 Symbol 11 “Consult Accompanying Documents” Warning Symbol, the Rated Supply DC Voltage/ Amperage, the Rated Supply Input Voltage/Frequency/ Amperage.	(DO-2) Labelling		

(UN-8) The user needs a device with appropriate labeling in compliance with applicable regulatory requirements. _____	(DI-23) The device labeling/IFU shall incorporate appropriate information on risks/hazards/ and mitigations as identified by the risk assessment. _____	(DO-2) Labelling _____		
(UN-9) The user needs a device which is powered by and communicates by "power over ethernet" (POE) hidden in mounting area (not exposed) _____	(DI-24) Power shall be provided to the device by POE _____	(DO-7) Assembly _____		
(UN-10) The user needs to be provided with a dated use life for the device. _____	(DI-28) The device shall have a use life of 1 year. _____	(DO-7) Assembly _____		
(UN-11) The user needs a device with appropriate packaging to ensure the device is protected during storage and transportation. _____	(DI-29) Device will be packaged so that device and included accessories, as applicable, are undamaged at point of use. _____	(DO-1) The packaging consists of a cardboard box with foam inserts _____		
(UN-11) The user needs a device with appropriate packaging to ensure the device is protected during storage and transportation. _____	(DI-30) The device shall be packaged within a Cardboard Box + Foam, alongside the power adapter, power cable, and instructions for use. _____	(DO-1) The packaging consists of a cardboard box with foam inserts _____		
(UN-11) The user needs a device with appropriate packaging to ensure the device is protected during storage and transportation. _____	(DI-31) The device packaging shall include an outer device label. _____	(DO-2) Labelling _____		
(UN-12) The user needs an electrically safe device compliance with IEC 60601-1, IEC 60601-1-2, and IEC 60601-1-11. _____	(DI-33) Device design will be compliant to IEC 60601-1 to ensure medical device electrical equipment safety _____	(DO-7) Assembly _____		
(UN-12) The user needs an electrically safe device compliance with IEC 60601-1, IEC 60601-1-2, and IEC 60601-1-11. _____	(DI-35) The device shall not incorporate external connections other than POE _____	(DO-7) Assembly _____		
(UN-12) The user needs an electrically safe device compliance with IEC 60601-1, IEC 60601-1-2, and IEC 60601-1-11. _____	(DI-37) The EM emissions from the device shall not exceed FCC standards. _____	(DO-7) Assembly _____		

(UN-12) The user needs an electrically safe device compliance with IEC 60601-1, IEC 60601-1-2, and IEC 60601-1-11.	(DI-38) The device shall be able to function under potential EM interference from other devices up to FCC Standards.	(DO-7) Assembly		
(UN-13) Users need a device that can indicate device errors to the user.	(DI-39) The firmware shall be able to automatically detect the following error states: Power Issues, Hardware Issues, SW Issues.	(DO-8) Software Architecture		
(UN-14) The user needs a device which can transmit data to a cloud platform	(DI-40) The firmware shall support internet connectivity.	(DO-8) Software Architecture		
(UN-14) The user needs a device which can transmit data to a cloud platform	(DI-41) The device shall be capable of firmware over the air updates	(DO-8) Software Architecture (DO-9) SDS-001 FOTA Updates Firmware over the air updates are deployed via AWS. These updates are manually initiated from the dashboard of the user interface Firmware is delivered as Debian packages based on an assigned tag for the firmware version. The firmware can only pull from a private repository. The OS provider provides the packages as Debian packages via apt. Updates are only cleared for our distribution after they have been internally validated.		
(UN-15) The firmware needs to be able detect if a patient is moving and their presence/absence from the field of view	(DI-42) The firmware shall determine the patient's presence/absence and their relative movement from the signals received by the radar transmitter and the IR camera or the visual camera	(DO-5) RGB Camera (DO-8) Software Architecture		
(UN-16) The user needs to be able to see if a patient has fallen	(DI-43) The firmware shall be able to determine if a patient has fallen and send an alert to the cloud system	(DO-8) Software Architecture		
(UN-17) The user needs to be able to determine whether the patient is in their cell	(DI-44) The cloud front end shall display a thermographic image of the patient	(DO-4) User Interface Wireframes		

(UN-18) The user needs to be able to determine when a patient has fallen or moved	(DI-45) The Cloud front end shall display the last time a patient fell	(DO-4) User Interface Wireframes		
(UN-18) The user needs to be able to determine when a patient has fallen or moved	(DI-46) The cloud front end shall display the last time a patient moved	(DO-4) User Interface Wireframes		
(UN-20) The user needs to be able to organize their list of patients	(DI-50) The cloud front end shall allow users to organize patients into groups.	(DO-4) User Interface Wireframes		
(UN-21) The user needs to be able to set up and configure devices	(DI-51) The cloud front end shall allow users to enable or disable to devices, view device issues, and edit associated information such as the name of the patient being monitored	(DO-4) User Interface Wireframes		
(UN-1) The user needs a device with a sensor that is able to continuously and accurately measure falls and the presence of a patient in their cell (UN-4) The user needs an unobtrusive and rugged enclosure that will protect all internal components from inmates and washing	(DI-52) Device shall incorporate a visual camera	(DO-5) RGB Camera		
(UN-22) The user needs the device to monitor the entire cell from the ceiling which is typically 10 ft tall and 10x10ft in size	(DI-53) The motion and fall detection capabilities will cover the entire 10x10ft room from an elevation of 10 ft, which is approximately a distance of at least 13 feet	(DO-4) User Interface Wireframes		
(UN-23) The user needs a device that limits the number of false alerts	(DI-54) The motion and fall detection system will have less than 1 false alarm per 4 hours using test data	(DO-8) Software Architecture		
(UN-1) The user needs a device with a sensor that is able to continuously and accurately measure falls and the presence of a patient in their cell	(DI-55) The Fall detection algorithm shall trigger an alert when a fall is detected, then immediately return to its default state after the alert has been acknowledged	(DO-8) Software Architecture		

(UN-1) The user needs a device with a sensor that is able to continuously and accurately measure falls and the presence of a patient in their cell	(DI-56) The system shall be able to distinguish between falls and other activities (such as exercise) that involve rapid vertical motion by checking that the patient remains on the ground for a certain period of time	(DO-8) Software Architecture		
(UN-1) The user needs a device with a sensor that is able to continuously and accurately measure falls and the presence of a patient in their cell	(DI-57) Immediately after detecting a fall, the Cloud UI will flag that the user fell at that specific time	(DO-4) User Interface Wireframes		
(UN-24) The user needs a Cybersecure device	(DI-58) The device shall not store identifiable information about patients	(DO-9) SDS-001 FOTA Updates Firmware over the air updates are deployed via AWS. These updates are manually initiated from the dashboard of the user interface Firmware is delivered as Debian packages based on an assigned tag for the firmware version. The firmware can only pull from a private repository. The OS provider provides the packages as Debian packages via apt. Updates are only cleared for our distribution after they have been internally validated. (DO-10) SDS-002 Privacy Measures The device does not store any identifiable information about the inmate. Once it is set up, it automatically begins monitoring everything in its field of view. It is not given any information about the subject it is monitoring. No RGB video is transferred off the device and the video is not stored on the device itself.		